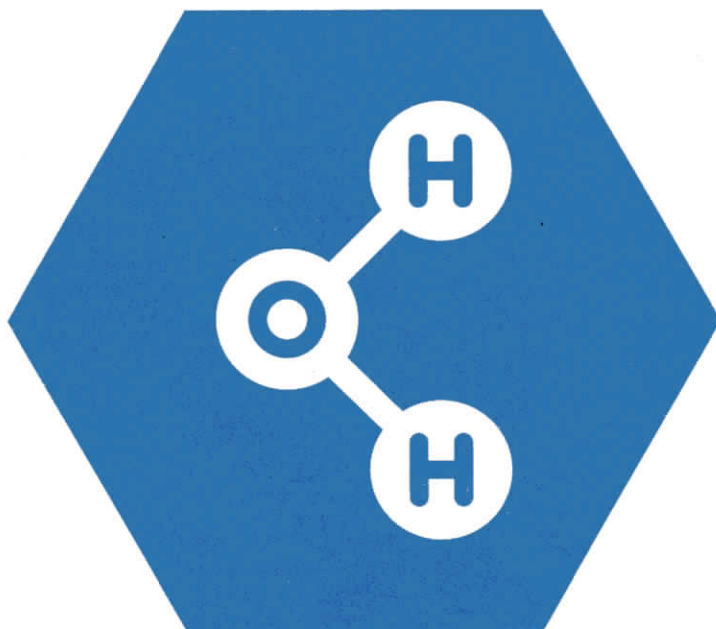




UNSW Global
THE UNIVERSITY OF NEW SOUTH WALES
SYDNEY • AUSTRALIA

PAPER A



2014 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.
Use a 2B or B pencil.
Do **NOT** use a pen.
Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

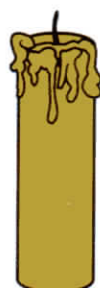
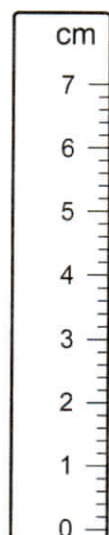
Mark only **ONE** answer for each question.
Your score will be the number of correct answers.
Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).
Use the information provided to choose the **BEST** answer from the four possible options.
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

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1. Milan measures the height of this candle with a ruler.

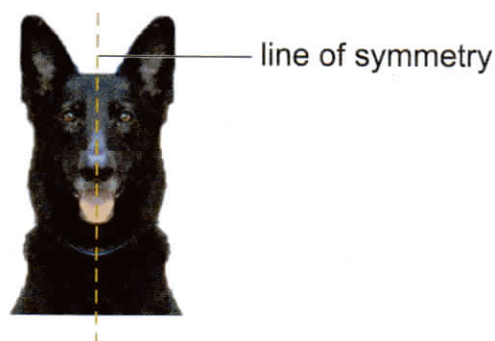


What is the height of the candle?

- (A) 7 cm
- (B) 6 cm
- (C) 4 cm
- (D) 1 cm

2. Many living things are symmetrical.

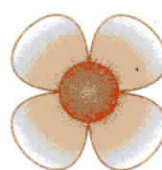
For example, this picture of a dog has one line of symmetry.



The pictures below show a crab and a flower.



crab



flower

Which option has the correct number of lines of symmetry for each picture?

	crab	flower
(A)	1	2
(B)	2	2
(C)	1	4
(D)	2	4

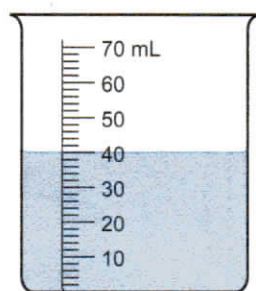
3. The table lists the features of some different shapes.

Shape	Colour	Size
square	orange	medium
circle	blue	large
rectangle	red	medium
square	orange	small
circle	yellow	small

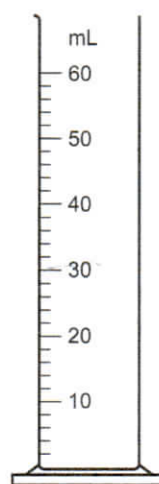
What colour is the small square shape?

- (A) orange (B) blue (C) red (D) yellow

4. Joe had two containers, X and Y.



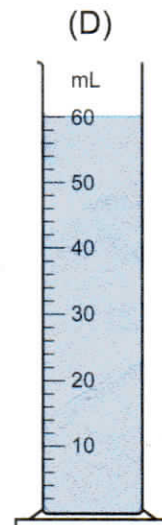
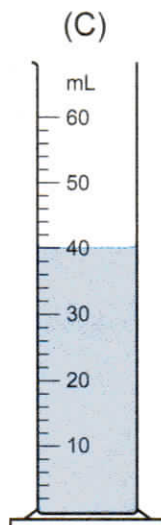
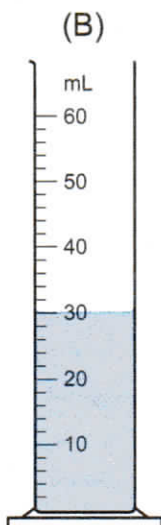
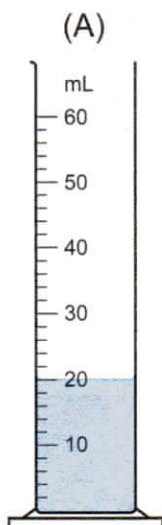
container X



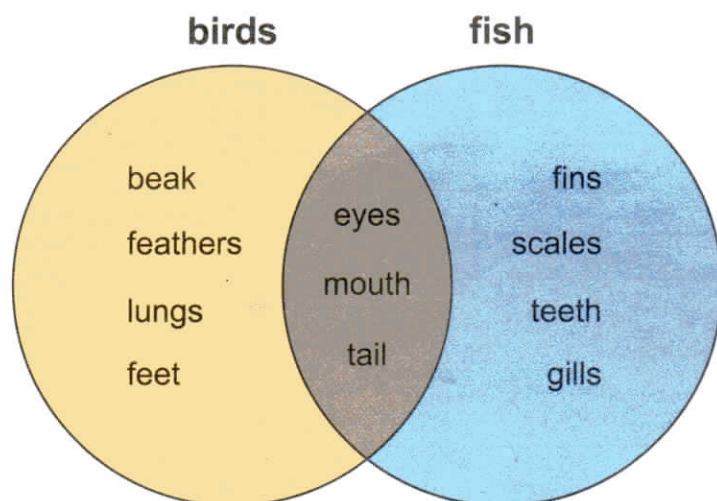
container Y

All the water from container X was poured into container Y.

Which diagram shows the water level in container Y?



5. The diagram compares the features of birds with the features of fish.

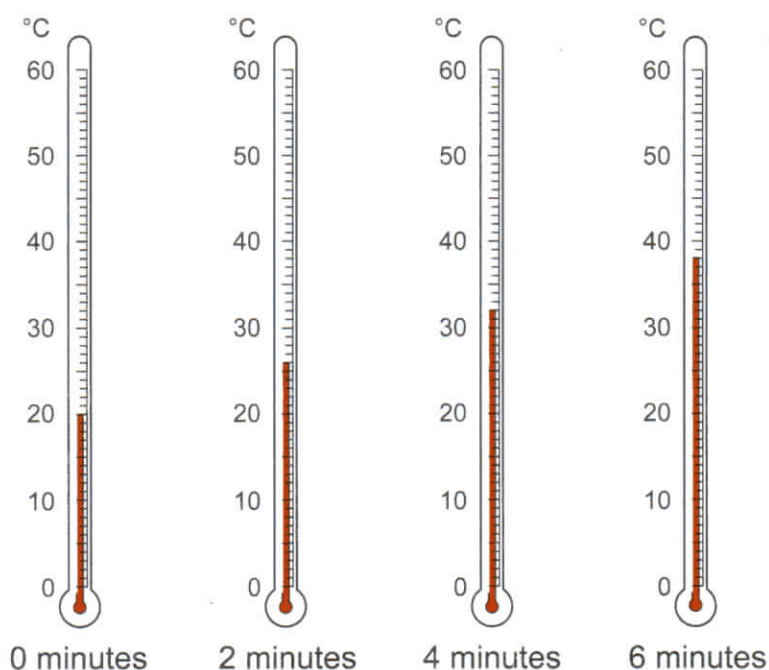


What are two features that birds and fish have in common?

- (A) mouth and tail
- (B) fins and teeth
- (C) teeth and eyes
- (D) feathers and lungs

6. Adam heated some water for eight minutes. He measured the water temperature using a thermometer.

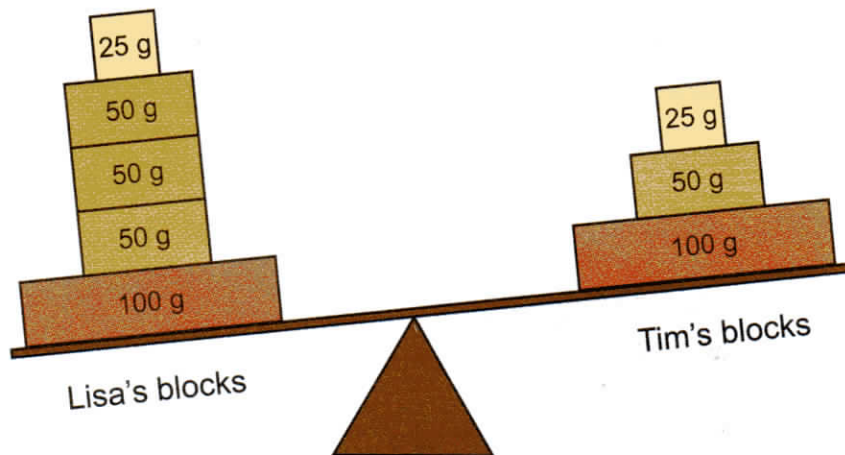
The thermometers show the temperature of the water every two minutes.



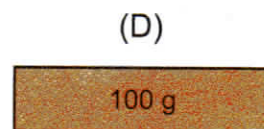
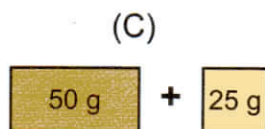
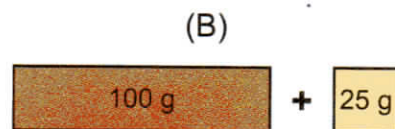
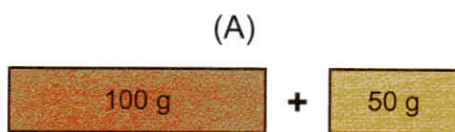
What will the temperature of the water be at 8 minutes?

- (A) 38 °C
- (B) 42 °C
- (C) 44 °C
- (D) 46 °C

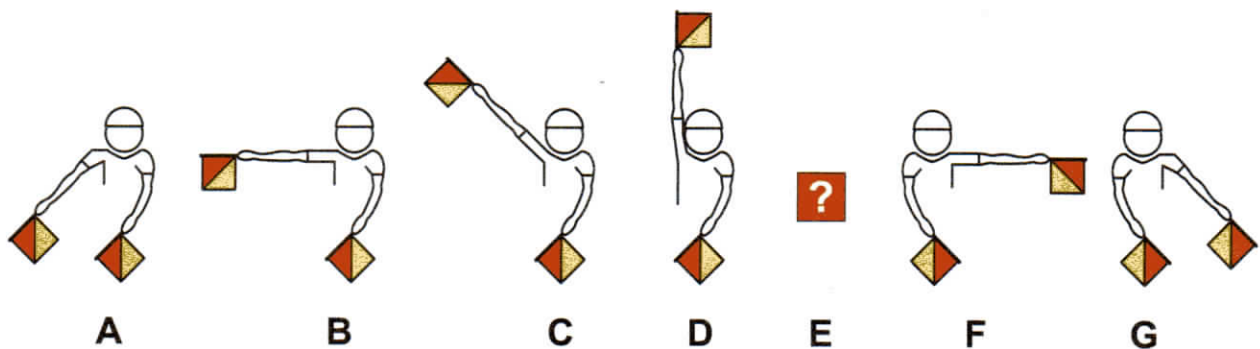
7. Lisa and Tim were adding blocks to a seesaw.



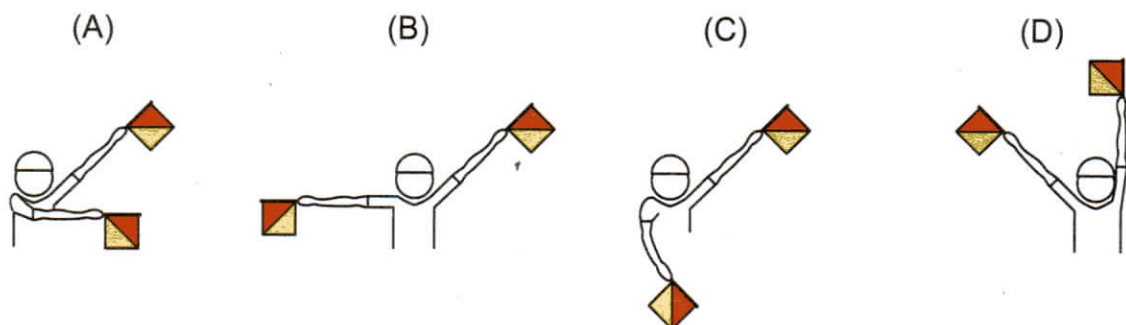
Which block(s) should Tim add to his side to balance the seesaw?



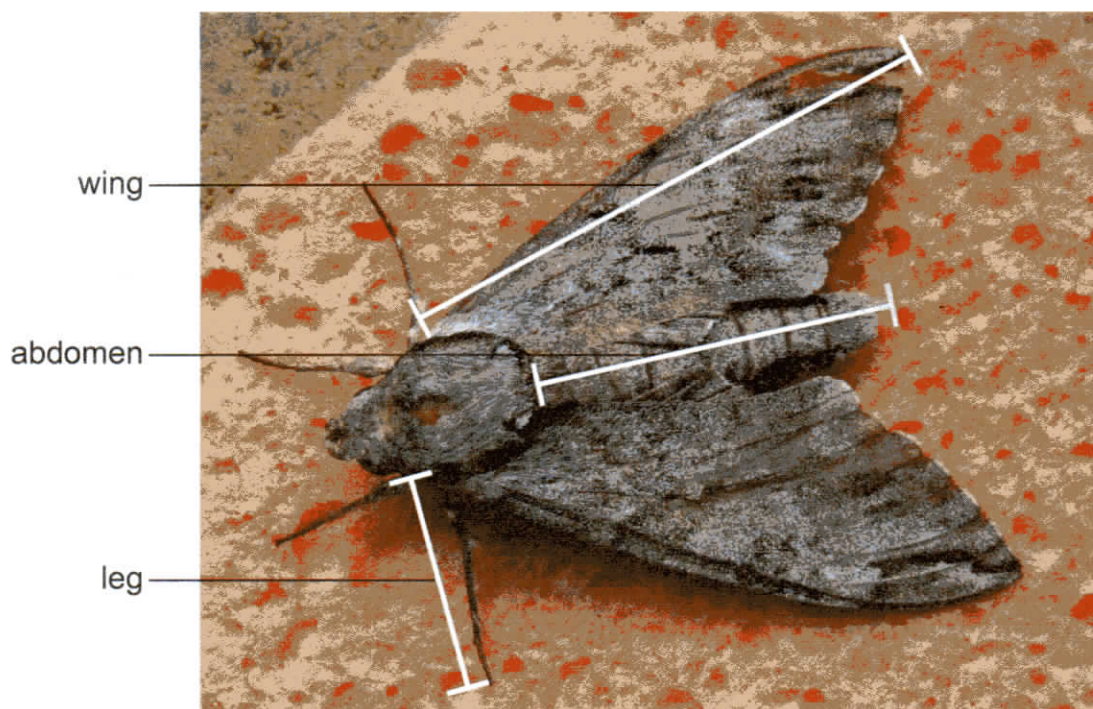
8. Flags can be used to send messages. The position of a pair of flags held by a person matches a letter of the alphabet. The positions of the flags follow a pattern.



Which of the following represents the letter E?



9. This moth landed on a sponge. Didi used a ruler to measure the length of some parts of its body.

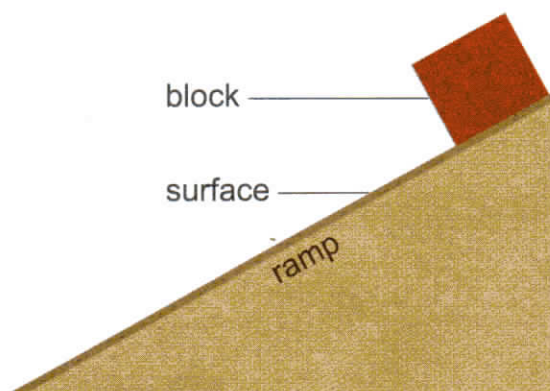


Which of the following statements is correct?

- (A) The leg, wing and abdomen are all the same length.
- (B) The abdomen is longer than the wing which is longer than the leg.
- (C) The leg is shorter than the wing which is longer than the abdomen.
- (D) The abdomen is shorter than the leg which is longer than the wing.

10. Friction is a force that slows down the movement of objects.

Some students recorded the time it took for one block to slide down four ramps. The ramps were the same, except each ramp had a different surface.



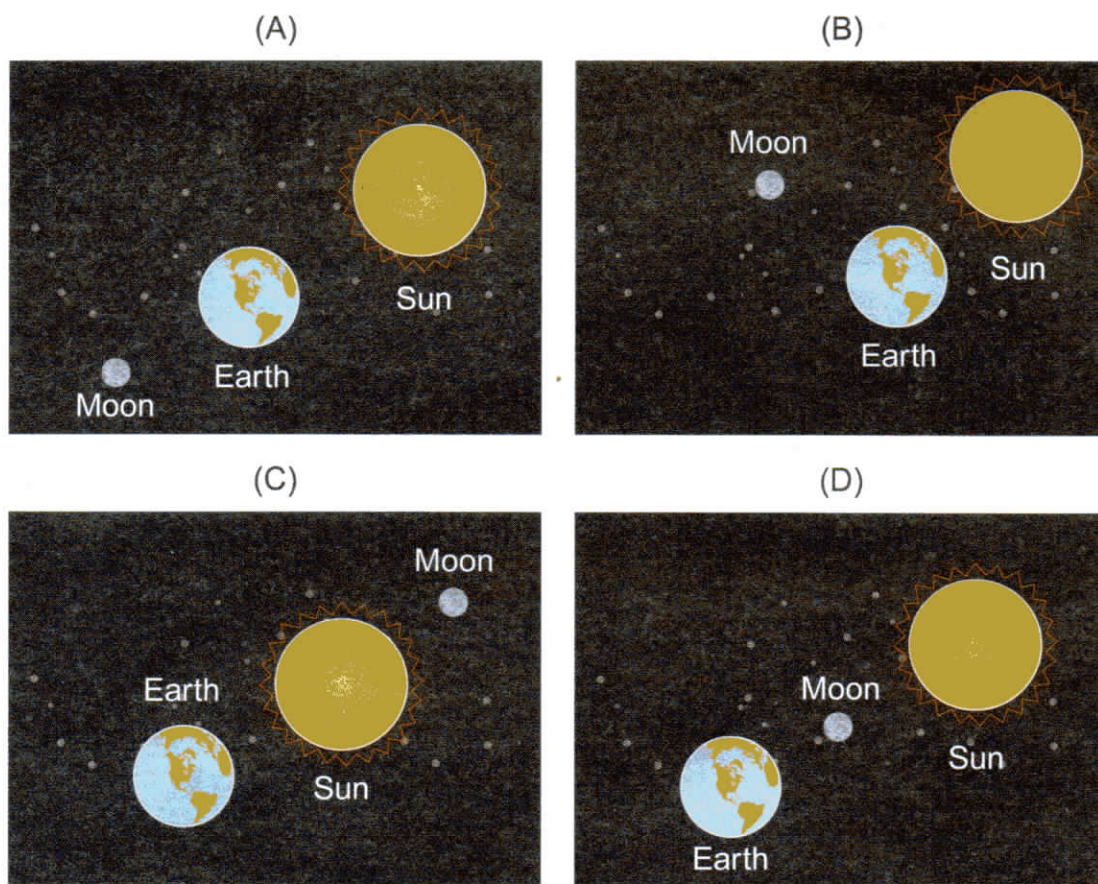
Surface	Time taken for block to slide down the ramp (seconds)
carpet	19
glass	5
wood	10
rubber	12

Which surface caused the most friction with the block?

- (A) carpet
- (B) glass
- (C) wood
- (D) rubber

11. A solar eclipse occurs when the Moon blocks some of the sunlight from reaching Earth.

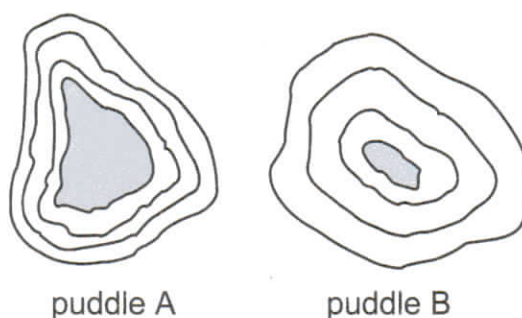
Which diagram shows a solar eclipse?



(NOT TO SCALE)

12. Sarah made two puddles on her floor using two different liquids. She used 100 mL of each liquid and poured them into separate puddles at the same time.

Sarah left the puddles to dry for 4 hours. Every hour she drew a line around what was left of each puddle as shown in the diagram.



puddle A

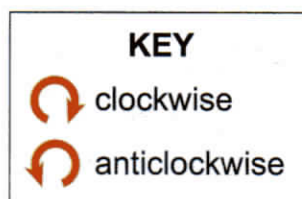
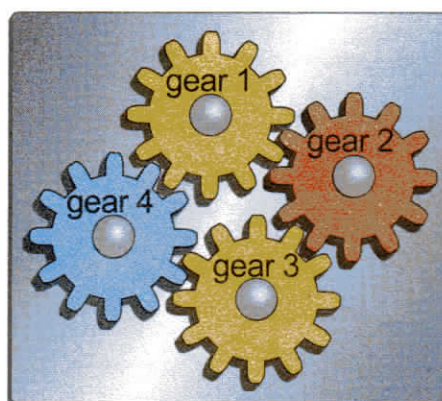
puddle B

What can Sarah conclude from her observations?

- (A) The liquid in puddle A was water.
(B) Puddle B had more liquid than puddle A.
(C) The liquid in puddle B stopped drying after 4 hours.
(D) The liquid in puddle A dried more slowly than the liquid in puddle B.

13. Gears are wheels with teeth.

The diagram shows four gears.



Gear 1 is rotating clockwise.

What is the direction of rotation of each of the other gears?

	Gear 2	Gear 3	Gear 4
(A)	clockwise	anticlockwise	clockwise
(B)	anticlockwise	clockwise	anticlockwise
(C)	anticlockwise	anticlockwise	clockwise
(D)	clockwise	clockwise	anticlockwise

14. Michelle counted how many weeds of different types she found in four areas of a nature reserve.

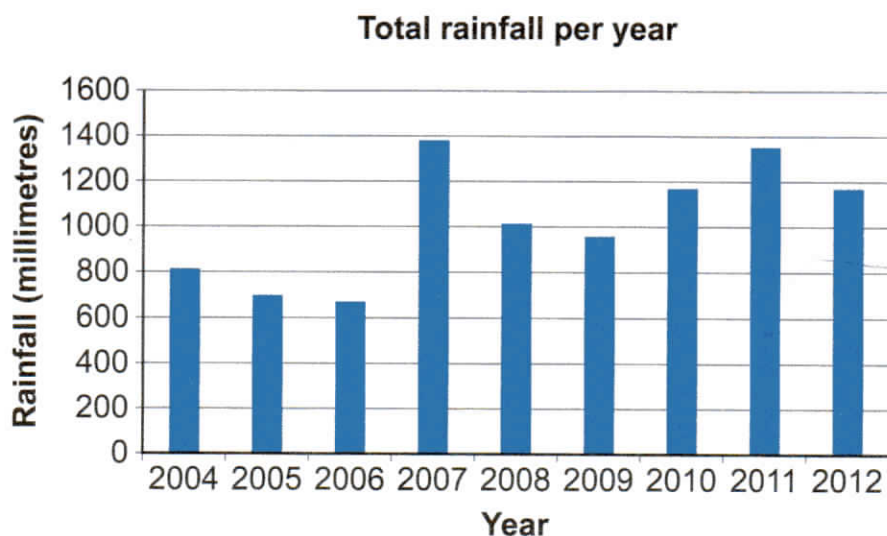
The table shows her results.

	Number of each weed		
Area	Dandelions	Thistles	Blackberries
1	5	7	0
2	3	7	9
3	6	3	13
4	0	12	0

Which statement about the weeds that Michelle found is correct?

- (A) She found dandelions in all four areas.
- (B) She found more blackberries than dandelions.
- (C) She found blackberries wherever there were thistles.
- (D) She found more thistles than dandelions in each area.

15. This graph shows the rainfall per year for a city.



Which years had a total rainfall less than 900 millimetres?

- | | |
|-------------------------|-------------------------|
| (A) 2004, 2005 and 2006 | (B) 2007, 2008 and 2009 |
| (C) 2004 and 2009 only | (D) 2005 and 2006 only |

16. The diagrams show four tiles, each with part of a plant drawn on it.



tile 1



tile 2



tile 3

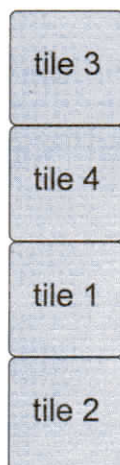


tile 4

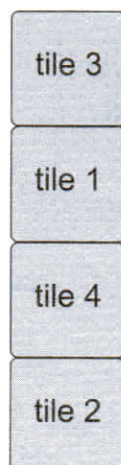
When the tiles are lined up correctly, the drawings show the complete plant.

How do the tiles have to be lined up to show the complete plant?

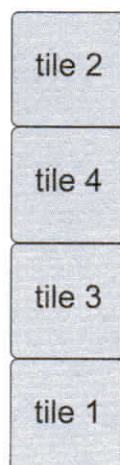
(A)



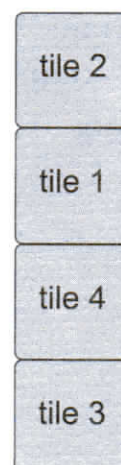
(B)



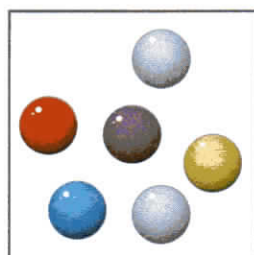
(C)



(D)



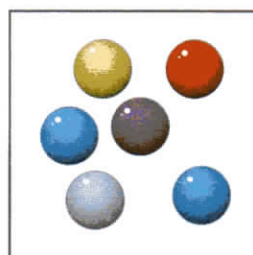
17. Lisa makes five groups of coloured marbles that follow a pattern.



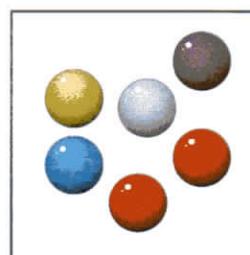
Group 1



Group 2



Group 3



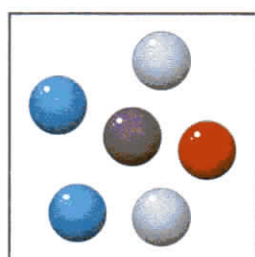
Group 4

Which of the following is the fifth group of marbles?

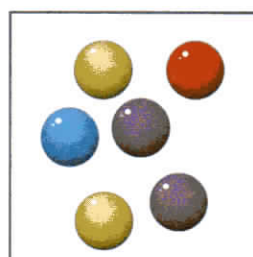
(A)



(B)



(C)



(D)



For questions 18 and 19 use the information below.

The table shows the number of days it takes for different seeds to germinate when planted in soil at different temperatures.

Days for seeds to germinate at different temperatures				
Seeds	10 °C	15 °C	20 °C	25 °C
onion	13	7	5	4
spinach	12	7	6	5
lettuce	7	4	3	2
cabbage	17	10	7	6
carrot	17	10	7	6
pea	14	9	8	6
tomato	43	14	8	6
corn	22	12	7	4
bean	–	30	17	6
cucumber	–	13	6	4

18. Which seed will take the longest time to germinate at 15 °C?

(A) onion (B) cabbage (C) tomato (D) bean

19. Which is the fastest germinating seed at any temperature?

(A) spinach (B) tomato (C) cucumber (D) lettuce

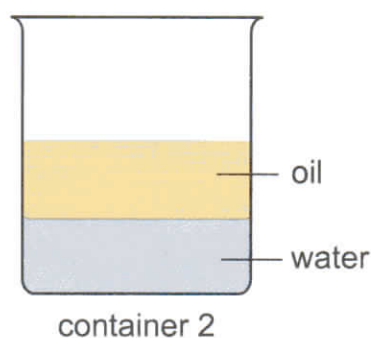
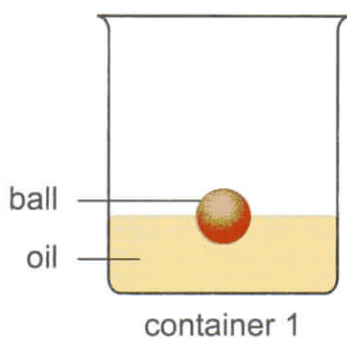
20. The table shows the times that the Sun rose and set during one week. The information for day 5 is missing.

Day	Sun rose at	Sun set at
1	5:50 am	5:56 pm
2	5:48 am	5:57 pm
3	5:46 am	5:58 pm
4	5:44 am	5:59 pm
5		
6	5:40 am	6:01 pm
7	5:38 am	6:02 pm

Based on the table, at what times did the Sun rise and set on day 5?

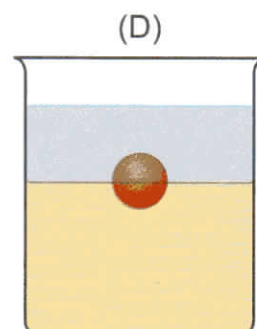
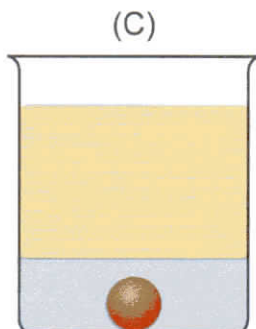
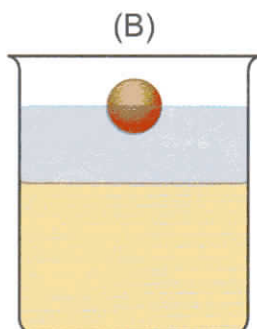
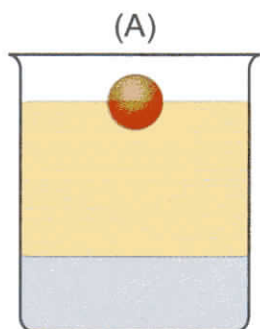
	Sun rose at	Sun set at
(A)	5:42 am	6:00 pm
(B)	5:42 am	5:58 pm
(C)	5:46 am	6:00 pm
(D)	5:46 am	5:58 pm

21. Container 1 had oil and a small rubber ball in it. Container 2 had water and oil in it.



Jane poured the contents of container 2 into container 1.

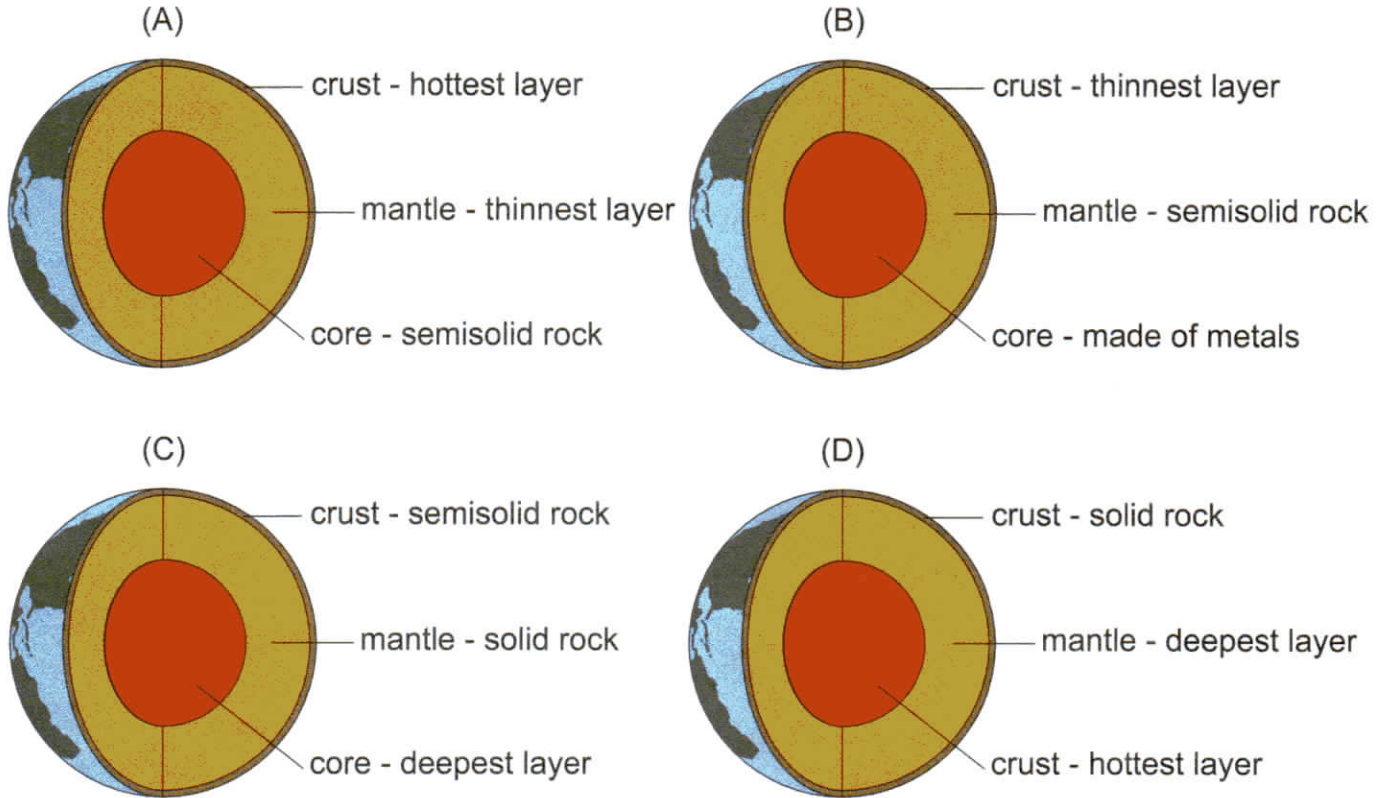
Which diagram shows what container 1 looks like now?



22. Earth is made up of three main layers: the crust, the mantle and the core.

- The crust is the thinnest layer and is solid rock. We live on the crust.
- The mantle is very hot and consists of mainly semisolid rock.
- The core is the deepest and hottest layer and is made of metals.

Which diagram correctly matches the information about the layers of Earth?



23. Matt was walking along the sand when he saw these footprints.



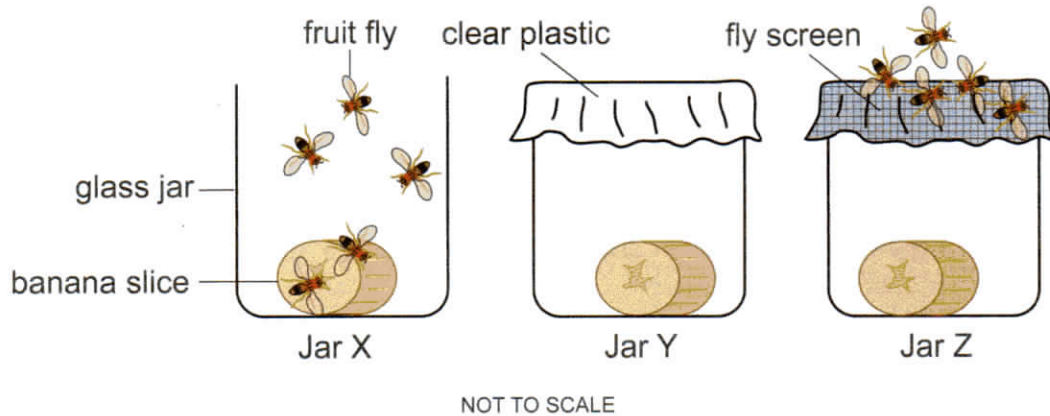
Which animal was the last to leave these footprints on the sand?

- (A) cat (B) bird (C) human (D) horse

For questions 24 and 25 use the information below.

Nathan wanted to find out how fruit flies find bananas.

Nathan cut three pieces from one banana and placed them into three identical glass jars. He covered two of the jars with different materials. He placed all three jars together on a table outside. The diagrams show what Nathan observed a short time later.



24. Based on these observations, how do fruit flies find bananas?

- | | |
|------------------------------|------------------------|
| (A) by sight | (B) by smell |
| (C) by either sight or smell | (D) by smell and taste |

25. Which condition did Nathan have to keep the same for each jar to make his investigation a fair test?

- | | |
|-------------------------------|----------------------------------|
| (A) the type of cover | (B) the type of fruit fly |
| (C) the number of fruit flies | (D) the size of the banana slice |

26. Earthworms usually live in soil where it is cool and moist and where there are air spaces between the soil particles.

After a day of heavy rain, Josh went outside and looked at his garden. He saw many earthworms on the surface of the wet soil.



What is the most likely reason these earthworms crawled out of the wet soil?

- (A) The soil was becoming too dry and they prefer moist soil.
- (B) There were too many earthworms living in the same area.
- (C) The earthworms come to the surface so they can breathe.
- (D) The earthworms need to keep their bodies wet.

27. Magnets have a north pole and a south pole. When magnets are placed near one another, they can either repel or attract one another as shown in the diagram.



Four students were given the challenge of joining four magnets together in a line so that they would remain joined.

Which line of magnets will join together?

- (A)    
- (B)    
- (C)    
- (D)    

28. Scientists made a new material with these features:

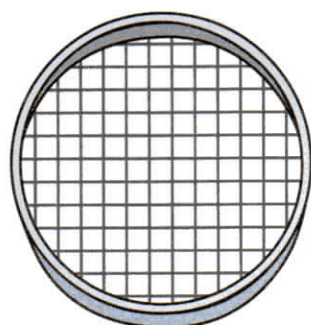
- waterproof
- cannot be bent
- does not burn
- can be heated up to 300 °C
- allows heat to pass through it.

Which of these is the most suitable product that could be made using this new material?

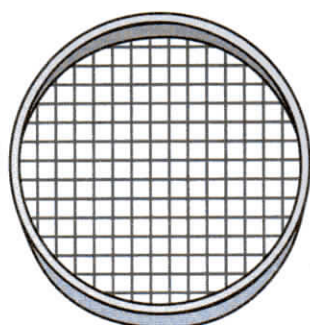
- (A) oven gloves (B) garden hose
(C) drinking glass (D) saucepan

For questions 29 and 30 use the information below.

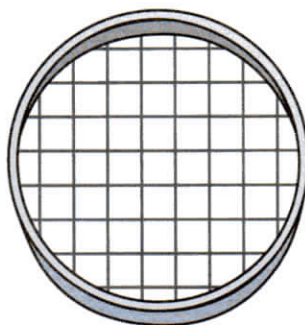
A sieve is used to separate particles of different sizes. Toby has four sieves and four foods.



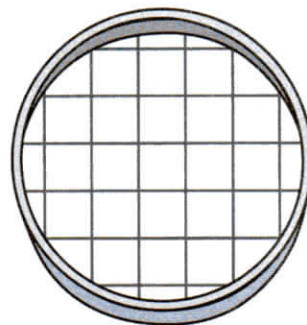
sieve W



sieve X



sieve Y



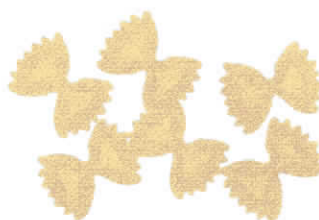
sieve Z



black-eye peas



rice



dried pasta



lentils

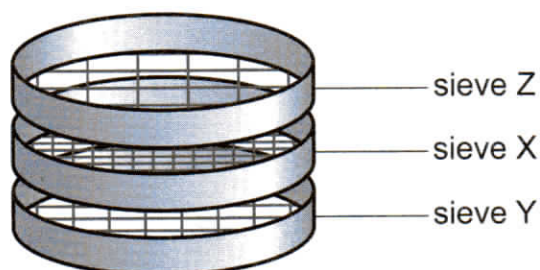
29. Which sieve will allow only two of these foods to pass through?

- (A) W (B) X (C) Y (D) Z

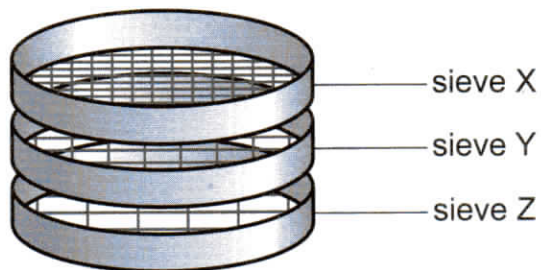
30. Toby only needs three sieves to separate the four foods. He places a mixture of the four foods into the top sieve and shakes the sieves.

Which combination of sieves will separate the four foods?

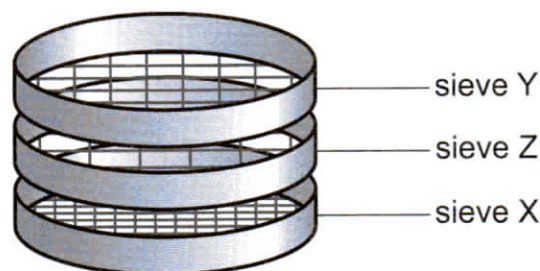
(A)



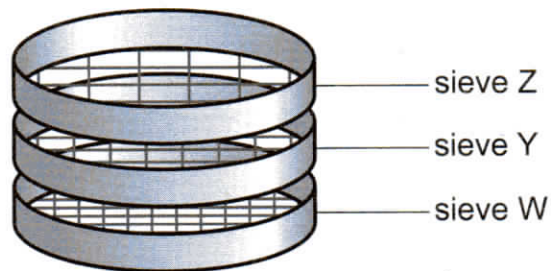
(B)



(C)



(D)



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Sources

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The following year levels should sit THIS Paper:

Australia	Year 3
Brunei	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific Region	Year 4
Singapore	Primary 2
South Africa	Grade 3



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S

2014 Science Answer Keys

ICAS

International Competitions
and Assessments for Schools

Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
1	C	A	B	B	B	D	C	A	B
2	C	C	B	D	C	D	D	B	D
3	A	B	A	B	C	B	B	D	D
4	C	C	C	D	D	B	B	B	D
5	A	B	C	D	A	C	C	B	A
6	C	C	D	C	A	A	C	B	B
7	D	B	D	A	B	C	C	A	D
8	C	B	C	B	C	B	B	A	B
9	C	D	C	B	B	C	C	D	B
10	A	C	A	C	C	D	D	B	A
11	D	D	A	D	B	B	C	B	D
12	D	A	B	C	C	B	D	C	D
13	B	D	C	D	D	C	B	C	A
14	B	D	B	B	D	D	C	C	C
15	A	C	A	D	C	A	B	A	A
16	B	D	B	C	D	B	C	B	C
17	A	C	A	B	A	C	A	B	C
18	D	A	A	C	B	A	B	D	C
19	D	B	B	A	B	D	D	A	D
20	A	C	C	B	C	B	D	C	A
21	A	D	A	B	D	B	C	C	C
22	B	B	D	A	B	C	B	D	A
23	A	B	B	A	D	C	D	C	C
24	B	A	C	B	C	B	B	C	C
25	D	D	A	D	C	B	D	B	D
26	C	D	A	B	C	B	C	C	A
27	B	A	A	C	A	A	D	A	B
28	D	B	B	C	A	C	B	D	D
29	A	A	D	A	A	A	C	D	C
30	D	D	D	A	B	D	D	D	B

(Please turn over)



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2014 Science Answer Keys

ICAS

International Competitions
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Question Number	Paper A	Paper B	Paper C	Paper D	Paper E	Paper F	Paper G	Paper H	Papers I & J
31			C	C	B	A	C	D	C
32			B	D	D	D	C	B	D
33			A	A	A	D	C	A	B
34			A	C	C	B	C	D	A
35			A	A	B	D	A	A	B
36			B	A	C	C	A	B	C
37			D	D	D	D	D	C	B
38			B	A	A	A	C	D	D
39			D	D	A	C	D	D	D
40			D	D	B	B	C	C	C
41					D	D	B	D	B
42					B	D	C	C	B
43					C	A	D	C	C
44					C	C	A	B	C
45					A	D	D	B	B

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